Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/14/2007
PI Name:	Fuller, Charles A. Ph.D.		
Project Title:	Artificial Gravity and Light as Countermeasures for Circadian Dysfunction in Altered Gravity Environments		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	06/01/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Funding reinstated to 06/01/2009, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Robinson, Edward L Ph.D. (University of California, Davis)		
Grant/Contract No.:	NNJ04HF44G		
Performance Goal No.:			
Performance Goal Text:			
	The Circadian Timing System (CTS) is an important temporal organizer controlling both the physiology and behavior of organisms. The CTS has been shown to be sensitive to altered gravity (AG) in both the microgravity of space flight and in hypergravity produced by centrifugation. Evidence from several species shows that AG produces changes in the period of the circadian clock (tau). Evidence from rodents also shows that AG can phase-shift and synchronize the circadian pacemaker. Further, sensitivity to light exposure, the most potent synchronizer of the CTS, is evidently impaired in AG, as shown by both molecular and physiological evidence. Thus, circadian regulation in AG may result from multiple interactions of G and light on the CTS. It is important to understand how AG alters the CTS response to light exposure to ensure that countermeasures for CTS dysfunction such as timed bright light are equally effective in AG. We will use AG produced by centrifugation to determine if the phase-shifting responses to light are impaired in		

Task Description:	AG. Countermeasures for circadian may differ from those seen in 1G. We will test a light countermeasure for circadian dysfunction, with defined spectral content, duration and schedule, to determine if its efficacy is affected by AG. Timed AG exposure by centrifugation is a proposed countermeasure for changes in cardiovascular, skeletal, muscle and other systems in space flight. Pulses of AG, however, can entrain and phase shift the CTS in rats. Thus the use of a timed AG countermeasure could potentially compromise CTS function, or conflict with circadian countermeasures, by inappropriately phase-shifting the CTS. It is important to understand how acute AG exposure at different times of day alters circadian phase and period in a diurnal primate. Impaired phase shifting by light would impact adjustment to new schedules as well as entrainment to either 24-h or non-24-h days. Conversely, if AG is capable of phase shifting or entraining the primate CTS, AG could also serve as a countermeasure for circadian dysfunction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program will characterize disturbances of sleep in rhesus monkeys. The rhesus is a well established model for human physiology in many disciplines and is showing promise as a model of human sleep disturbances. This study will directly relate to issues of sleep and circadian rhythm disturbance as a result of occupational factors such as shift work or other irregular schedules and jet lag. By establishing counter-measures which improve sleep and performance in both hypergravity and normal gravity, we will be advancing what is known about potential treatment of sleep problems on earth. Further, optimization of countermeasures for space flight will likely involve controlled light exposure to help ensure adequate sleep. This will need to be based on an understanding of how the primate visual system responds to light, in particular to spectral content and intensity. Recent findings of that non-visual photoreception is linked to the CTS and likely to sleep consolidation as well underscores the importance of quantitative development of specific light countermeasures. Understanding sleep disturbances and the responses of the CTS, including sleep and performance to to altered gravity is also important to extending the ability of humans to function effectively in stressful terrestrial environments.
Task Progress:	This research program has made significant progress in examination of the mechanisms underlying the response of the circadian timing system (CTS) of primates to altered gravity. We have initiated new studies on the interaction of altered gravity and light sensitivity on the CTS. We are examining changes in circadian rhythms, sleep and performance to controlled lighting environments, with and without the perturbation of hypergravity produced by centrifugation. Several improvements have been made to the telemetry system and to the centrifuge. In addition we have collaborated on a revision of the Psychomotor Test System for performance metrics, which also serves as enrichment by providing an operant task to the animals. We have recently employed a short period light-dark cycle to study the effects of mild sleep restriction similar to that experienced by astronauts in space flight. The results to date from 1G studies have shown that sleep restriction by this method results in sleep fragmentation and altered frequency of specific sleep stages. Daytime occurrence of sleep is more prevalent, notably deep slow wave sleep. Decreased total REM sleep is also seen. When compared with changes in hypergravity, these findings will help us to define requirements for countermeasures to be employed in space flight. We have also compared sleep between genders in rhesus monkeys. Gender differences in sleep in humans are known and differences seen in rhesus monkeys support their use as a model for human sleep. More sleep was seen over the 24-h day in male rhesus. Males also showed more time in deep slow wave sleep, thought to represent the recuperative stages of sleep, as well as more time in REM sleep as compared with females. Timing of sleep was on average different between genders. Females tended to sleep and waken later than males relative to the light-dark cycle. However both genders exhibited delayed onset of deep slow wave sleep, marked by delta EEG, than humans. Although timing of deep sleep differed with humans, the rhesus exhibited similar major features with human sleep, validating their use as a model system. These features included a consolidated nocturnal sleep period, decreased deep slow wave sleep over the course of the night, and increased REM sleep during the night. We are currently evaluating the changes to sleep in response to altered gravity, including differences in response between genders.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Hsieh KC, Robinson EL, Fuller CA. "Effects of short light-dark cycles on sleep in rhesus monkeys." 19th Annual Meeting of the Associated Professional Sleep Societies (APSS), Denver, CO, June 2005. 19th Annual Meeting of the Associated Professional Sleep Societies (APSS), June 2005. , Jun-2005
Abstracts for Journals and Proceedings	Robinson EL, Hoban-Higgins TM, Fuller CA. "Artificial Gravity and Light as Countermeasures for Circadian Dysfunction in Altered Gravity Environments." Bioastronautics Investigators Workshop, Galveston, TX, January 2005. Bioastronautics Investigators Workshop, Galveston, TX, Jan-2005. , Jan-2005
Articles in Peer-reviewed Journals	Hoban-Higgins TM, Robinson EL, Fuller CA. "Primates in space flight." Adv Space Biol Med. 2005;10:303-25. Review. PMID: 16101112 (Experimentation with Animal Models in Space) , Jun-2005
Dissertations and Theses	Hsieh J. K.-C. "Homeostatic and Circadian Regulation of Sleep in Rhesus Monkeys." Molecular, Cellular and Integrative Physiology, University of California, Davis, September 2005. , Sep-2005